

Generation Interconnection System Impact Study Report

For Queue Project Q18-1052

Emory Grove 138 kV

March 19, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1052. The Interconnection Customer proposes a 187.4 MW solar photovoltaic facility interconnecting at the Emory Grove 138 kV substation in the Maryland area. The request was received on August 31, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 341 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$33,852,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Rockville Substation / Phillips Pump 1 Substation 138.0 kV	Tramway / Rossmoor S 138.0 kV L	161.1	162	30.8
Trumann / Rossmoor South 138.0 kV line	Tramway / Scarcello 138.0 kV	153.9	229	25.7
Scarcello / Rockville 138.0 kV L	Tramway / Rossmoor S 138.0 kV Line	147.5	194	30.8
Tamway / Scarcello 138.0 kV	Tramway / Rossmoor S Substation 138.0 kV	143.9	245	30.8
Bondurant / Montgomery Village 138.0 kV Ckt 1	Montgomery Village Substation / Bondurant 138.0 kV Ckt 2	142.7	361	36.3
Bondurant / Tramway 138.0 kV Line 1	Tramway / Bondurant 138.0 kV Cir. 2	138.2	360	36.1
Bondurant / Tramway 138.0 kV Cir. 2	Tramway Substation / Bondurant 138.0 kV #1	135.0	368	36.1
Montgomery Village / Emory Grove 138.0 kV	Air Force Base 345 kV/138 kV Autotransformer	130.5	177	73.2
Bondurant / Montgomery Village 138.0 kV Ckt 2	Montgomery Village Substation / Bondurant 138.0 kV Circuit 1	130.4	395	36.3
Tramway / Rossmoor South 138.0 kV	Tramway / Scarcello 138.0 kV L	129.8	313	25.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rockville Substation / Phillips Pump 1 138.0 kV	Rebuild line with higher-capacity conductor	\$2,181,000	100% (\$2,181,000)
Rossmoor South / Trumann 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,863,000	100% (\$5,863,000)
Rockville / Scarcello Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$997,000	100% (\$997,000)
Scarcello / Tramway 138.0 kV	Rebuild line with higher-capacity conductor	\$1,173,000	100% (\$1,173,000)
Bondurant / Montgomery Village 138.0 kV Ckt 1	Rebuild line with higher-capacity conductor	\$6,990,000	100% (\$6,990,000)
Bondurant / Tramway 138.0 kV Line 1	Rebuild line with higher-capacity conductor	\$2,442,000	100% (\$2,442,000)
Tramway / Bondurant 138.0 kV	Rebuild line with higher-capacity conductor	\$2,279,000	100% (\$2,279,000)
Emory Grove / Montgomery Village Substation 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,257,000	100% (\$1,257,000)
Montgomery Village / Bondurant 138.0 kV #2	Rebuild line with higher-capacity conductor	\$9,108,000	100% (\$9,108,000)
Tramway / Rossmoor South 138.0 kV	Rebuild line with higher-capacity conductor	\$1,562,000	100% (\$1,562,000)
	Total allocated to project		\$33,852,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.